Supplementary Material - The Incidence of Multimorbidity and Patterns in Accumulation of Chronic Conditions: A Systematic Review

Online Supplemental Table S1. PRISMA checklist.

PRISMA 2020 Main Checklist

Topic	No.	Item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Title page
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist	End of this checklist
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Section 1
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Section 1
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Section 2.2
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Section 2.1
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Online Supplemental Table S2
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Section 2.2
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Section 2.4
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Section 2.4

Topic	No.	Item	Location where item is reported
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Section 2.4
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Section 2.3, Online Supplemental Table S3
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Section 2.4
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item 5)).	Section 2.3, Section 2.4
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Section 2.4
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Section 2.4
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Section 2.4
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	N/A (meta-analyses not performed)
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	N/A (meta-analyses not performed)
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Not performed
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	N/A (meta-analysis not performed)
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Section 3.1, Figure 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Section 3.1
Study characteristics	17	Cite each included study and present its characteristics.	Table 1, Table 2

Topic	No.	Item	Location where item is reported
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Online Supplemental Table S4, Online Supplemental Table S5
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Table 3, Table 4
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Section 3.2
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Section 3.3, Section 3.4, Section 3.5
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Section 3
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Not performed
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Online Supplemental Table S4, Online Supplemental Table S5
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	N/A (meta-analysis not performed)
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Section 4
	23b	Discuss any limitations of the evidence included in the review.	Section 4
	23c	Discuss any limitations of the review processes used.	Section 4
	23d	Discuss implications of the results for practice, policy, and future research.	Section 4
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Section 2
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Section 2
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Available at PROSPERO registration.
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Title page

Topic	No.	Item	Location where item is reported
Competing interests	26	Declare any competing interests of review authors.	Title page
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Not made publicly available

PRISMA Abstract Checklist

Topic	No.	Item	Reported?
TITLE			
Title	1	Identify the report as a systematic review.	Yes
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	Yes
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	No
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	Yes
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	Yes
Synthesis of results	6	Specify the methods used to present and synthesize results.	Yes
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	Yes
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	Yes
DISCUSSION			
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	Yes
Interpretation	10	Provide a general interpretation of the results and important implications.	Yes

Topic	No.	Item	Reported?
OTHER			
Funding	11	Specify the primary source of funding for the review.	No
Registration	12	Provide the register name and registration number.	No

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. MetaArXiv. 2020, September 14. DOI: 10.31222/osf.io/v7gm2. For more information, visit: www.prisma-statement.org

Online Supplemental Table S2. Database search strategy.

Interface	Datasets	Search Strategy	Limits	Number of Articles Retrieved	Final no. Exported
OVID	Medline	1. multimorbidity.tw. 2. multi-morbidity.tw. 3. multiple morbidities.tw. 4. Multimorbidity/ 5. "multiple chronic conditions".tw. OR Multiple Chronic Conditions/ 6. "multiple chronic illnesses".tw. 7. multimorbidities.tw. 8. multi-morbidities.tw. 9. multiple-morbidities.tw. 10. "multiple chronic diseases".tw. 11. Incidence/ 12. inciden*.tw. 13. accumulat*.tw. 14. longitudinal.tw. 15. epidemiology.tw. 16. trend.tw. 17. trajector*.tw. 18. acquire*.tw. 19. acquisition.tw. 20. predict*.tw. 21. 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 22. 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 23. 21 AND 22	N/A	1. 4114 2. 545 3. 293 4. 1101 5. 1508 6. 111 7. 195 8. 87 9. 293 10. 438 11. 267969 12. 904255 13. 571995 14. 254612 15. 153796 16. 231093 17. 76979 18. 231093 19. 155011 20. 1633756 21. 3968428 22. 6982 23. 1659	1659
OVID	Embase	1. multiple chronic conditions/ 2. multimorbidity.tw. 3. multi-morbidity.tw. 4. multiple morbidities.tw. 5. multiple chronic conditions.tw. 6. multiple chronic illnesses.tw. 7. Multimorbidities.tw.	N/A	1. 3851 2. 5305 3. 921 4. 406 5. 1444 6. 157 7. 263	1681

		8. multi-morbidities.tw. 9. multiple-morbidities.tw 10. multiple chronic diseases.tw 11. Incidence/ 12. inciden*.tw. 13. accumulat*.tw. 14. longitudinal.tw. 15. epidemiology.tw. 16. trend.tw. 17. trajector*.tw. 18. acquire*.tw. 19. acquisition.tw. 20. predict*.tw. 21. 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 22. 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 24. 21 AND 22		8. 128 9. 406 10. 579 11. 433512 12. 1370178 13. 720694 14. 354614 15. 205425 16. 358072 17. 92829 18. 457341 19. 205814 20. 2233258 21. 9650 22. 5466464 24. 1681	
EBSCO	CINAHL	1. “multimorbidity” 2. “multi-morbidity” 3. “multiple morbidities” 4. "multiple chronic conditions" 5. "multiple chronic illnesses" 6. “multimorbidities” 7. “multi-morbidities” 8. “multiple-morbidities” 9. "multiple chronic diseases" 10. (MH "Incidence") OR "Incidence" 11. “accumulat*” 12. “Longitudinal” 13. “Epidemiology” 14. “Trend” 15. “trajector*” 16. “Acquire*” 17. “Acquisition” 18. “predict*”	N/A	1. 2188 2. 326 3. 127 4. 819 5. 77 6. 82 7. 60 8. 127 9. 203 10. 195,522 11. 32,580 12. 89,322 13. 438,805 14. 54,580 15. 20,745 16. 73,702 17. 43,063 18. 392,313 19. 3,651 20. 1,105,943	1236

		19. 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 20. 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 25. 19 AND 20		25. 1236	
Cochrane	Cochrane Library	1. “multimorbidity” 2. “multi-morbidity” 3. “multiple morbidities” 4. "multiple chronic conditions" 5. "multiple chronic illnesses" 6. “multimorbidities” 7. “multi-morbidities” 8. “multiple-morbidities” 9. "multiple chronic diseases" 10. "Incidence" 11. “accumulat*” 12. “Longitudinal” 13. “Epidemiology” 14. “Trend” 15. “trajector*” 16. “Acquire*” 17. “Acquisition” 18. “predict*” 19. 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 20. 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 21. 19 AND 20 26. 3	N/A	26. N/A	3

Online Supplemental Table S3. Clarification of NOS guideline definitions as applied by study authors; includes definitions for NOS categories and numbered items that required the application of thresholds decided by study authors. All items not requiring clarification were evaluated using standard definitions as stated by the NOS grading manual.

COHORT STUDIES			
Criteria, category (numbered item)[a]	**Category Grade**	**Star Allocation**	**No Star Allocation**
S(1). Representativeness of exposed cohort	*A) Truly representative*	Includes all ages and both sexes	Includes only persons of a specific age or sex
	B) Somewhat representative	Restricted age, sex, and/or health care setting	Persons with specific medical conditions
	C) Selected group of people	People of specific profession, group, or sharing specific condition(s)	Not applicable
S(2). Selection of non-exposed cohort		Non-exposed cohort drawn from the same community as exposed cohort	Non-exposed cohort drawn from different source(s) or no description of course provided
S(3). Ascertainment of exposure		Data sourced from secure medical records and/or structured interview(s)	Data sourced using self-report via questionnaire(s) and/or survey(s), and/or no description provided
S(4). Demonstration of outcome		Multimorbidity (outcome) was not present in participant(s) at the start of study	Multimorbidity (outcome) was present in participant(s) at the start of study
C(1). Comparability	*A) Most important factor*	Analysis adjusted for age	Analysis adjusted for age

	B) Second most important factor(s)	Analysis adjusted for additional factors other than age and/or age group (including sex and sociodemographic factors)	Analysis did not adjust for additional factors other than age and/or age group (including sex and sociodemographic factors)
O(1). Assessment of outcome		Clinical or physical assessment and/or medical record linkage	Self-report via questionnaire(s) or survey(s) and/or no statement
O(2). Adequacy of follow-up duration		≥ 5 years	< 5 years
O(3). Adequacy of follow-up rate		≥ 70% or less with explanation and/or cause of loss provided	< 70% follow-up rate with no explanation

CASE-CONTROL STUDIES			
Criteria, category (numbered item)*		**Star Allocation**	**No Star Allocation**
S(1). Case definition adequacy		No clarification required	
S(2). Case representativeness		No clarification required	
S(3). Selection of controls		No clarification required	
S(4). Definition of controls		No clarification required	
C(1). Comparability	*A) Most important factor*	Analysis adjusted for age	Analysis adjusted for age
	B) Second most important factor(s)	Analysis adjusted for additional factors other than age and/or age group (including sex and sociodemographic factors)	Analysis did not adjust for additional factors other than age and/or age group (including sex and sociodemographic factors)
E(1). Ascertainment of exposure		No clarification required	

E(2). Same method of ascertainment for cases and controls	No clarification required
E(3). Non-Response rate	No clarification required

[a] S, selection; C, comparability**;** O, outcome; E, exposure

Online Supplemental Table S4. Quality assessment of observational studies using the Newcastle-Ottawa Scale (NOS).

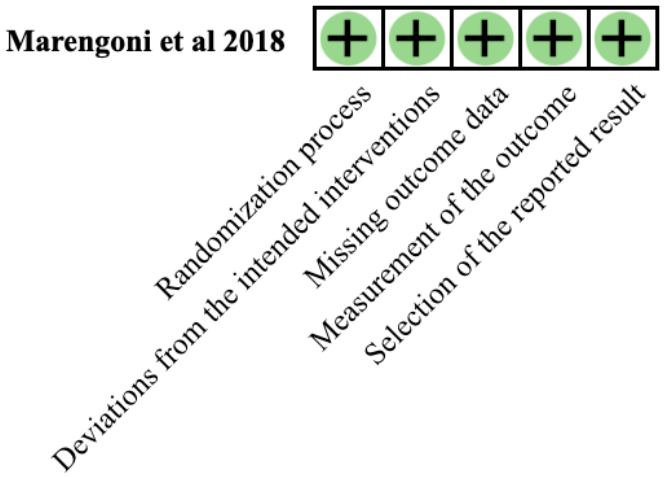
Study, year	Focus	Study design	Selection[a]	Comparability[a]	Exposure/ outcome[a]	Total score	AHRQ Standard
Aminisani et al, 2019	Incidence	Cohort	⋆ ⋆ ⋆	⋆ ⋆	⋆ ⋆	7	Good
Ashworth et al, 2019	Accumulation	Cohort	⋆ ⋆ ⋆ ⋆	⋆ ⋆	⋆	7	Poor
Bjur et al, 2019	Incidence	Cohort	⋆ ⋆ ⋆ ⋆	⋆ ⋆	⋆ ⋆	8	Good
Castilho et al, 2019	Incidence	Cohort	⋆ ⋆ ⋆	⋆ ⋆	⋆ ⋆	7	Good
Castilho et al, 2020	Incidence	Cohort	⋆ ⋆ ⋆	⋆ ⋆	⋆ ⋆	7	Good
Chan et al, 2019	Incidence	Cohort	⋆ ⋆ ⋆ ⋆	⋆ ⋆	⋆ ⋆ ⋆	9	Good
Demirchyan et al, 2013	Incidence	Cohort	⋆ ⋆	⋆	⋆	4	Poor
Dhalwani et al, 2017	Incidence	Cohort	⋆ ⋆ ⋆	⋆ ⋆	⋆	6	Poor
Dugravot et al, 2020	Incidence	Cohort	⋆ ⋆ ⋆	⋆ ⋆	⋆ ⋆ ⋆	8	Good
Freisling et al, 2020	Incidence & Accumulation	Cohort	⋆ ⋆ ⋆ ⋆	⋆ ⋆	⋆ ⋆	8	Good
Guaraldi et al, 2015	Incidence	Cohort	⋆ ⋆ ⋆	⋆ ⋆	⋆ ⋆	7	Good
Haug et al, 2020	Accumulation	Cohort	⋆ ⋆ ⋆ ⋆	⋆ ⋆	⋆ ⋆	8	Good
Hohmann et al, 2019	Incidence	Cohort	⋆ ⋆ ⋆	⋆ ⋆	⋆ ⋆	7	Good

Hussin et al, 2018	Incidence	Cohort	★ ★ ★ ★	★ ★	★	7	Poor
Jørgensen et al, 2020	Accumulation	Case-control	★	★	★ ★	4	Poor
Kivimäki et al, 2017	Incidence	Cohort	★ ★ ★ ★	★ ★	★ ★	8	Good
Lau et al, 2019	Incidence	Cohort	★ ★ ★ ★	-	★ ★	6	Poor
Licher et al, 2019	Incidence	Cohort	★ ★ ★ ★	★ ★	★ ★	8	Good
Melis et al, 2014	Incidence	Cohort	★ ★ ★ ★	★ ★	★ ★	8	Good
Mounce et al, 2018	Incidence	Cohort	★ ★ ★ ★	★ ★	★	7	Poor
Niedzwiedz et al, 2019	Incidence	Cohort	★ ★ ★ ★	★ ★	★ ★	8	Good
Ryan et al, 2018	Incidence	Cohort	★ ★ ★ ★	★ ★	★ ★	8	Good
Salloum et al, 2018	Incidence	Cohort	★ ★ ★	★ ★	★ ★	7	Good
Singh-Manoux et al, 2018	Incidence	Cohort	★ ★ ★ ★	★ ★	★ ★ ★	9	Good
St. Sauver et al, 2015	Incidence	Cohort	★ ★ ★ ★	★ ★	★ ★	8	Good
Strauss et al, 2014	Accumulation	Cohort	★ ★ ★	★ ★	★ ★	7	Good
Tomasdottir et al, 2016	Incidence	Cohort	★ ★ ★	★ ★	★	6	Poor
Ungprasert et al, 2017	Incidence & Accumulation	Case-control	★ ★ ★ ★	★ ★	★ ★ ★	9	Good

van den Akker et al, 1998	Incidence	Cohort	⋆ ⋆ ⋆ ⋆	⋆ ⋆	⋆	7	Poor
Wikström et al, 2015	Incidence	Cohort	⋆ ⋆ ⋆	⋆ ⋆	⋆ ⋆ ⋆	8	Good
Wilson-Genderson et al, 2017	Incidence	Cohort	⋆ ⋆ ⋆ ⋆	⋆ ⋆	⋆ ⋆	8	Good
Xu et al, 2018	Incidence	Cohort	⋆ ⋆ ⋆	⋆ ⋆	⋆ ⋆ ⋆	8	Good
Xu et al, 2019	Incidence	Cohort	⋆ ⋆ ⋆	⋆ ⋆	⋆ ⋆	7	Good
Xu et al, 2020	Incidence	Cohort	⋆ ⋆ ⋆	⋆ ⋆	⋆ ⋆	7	Good
Xu, Mishra & Jones, 2019	Incidence	Case-control	⋆ ⋆ ⋆	⋆ ⋆	⋆ ⋆	7	Good

[a] Each study could be awarded a minimum of 0 stars and a maximum of 4 stars for the *Selection* category, 2 for *Comparability,* and 3 for *Outcome,* totalling to a minimum of 0 and a maximum of 9 total stars.

Online Supplemental Table S5. Quality assessment of randomized control trials using the Cochrane risk of bias tool (RoB2).



Online Supplemental Table S6. Study definitions of multimorbidity and conditions included.

Study, year of publication	Definition of multimorbidity	Number of Individual Conditions and/or Condition Groups Included	Specific Conditions Included
Aminisani et al, 2019	Presence of 2+ included chronic diseases	17	Heart diseases, DM, stroke, COPD, DEP, AR, epilepsy, Parkinson disease, migraine headache, Alzheimer's disease/dementia osteoporosis, hip/knee replacement, asthma, chronic liver conditions (cirrhosis), cancer, anxiety, other mental diseases
Ashworth et al, 2019	Co-occurrence of 3+ included chronic diseases	12	DM, stroke, CHD, HF, AFIB, COPD, CKD, DEP, chronic pain, dementia, serious mental illness, morbid obesity
Bjur et al, 2019	Co-occurrence of 2+ included chronic conditions	31	Brain and spinal cord malformations, mental retardation, central nervous system degeneration and disease, infantile cerebral palsy, muscular dystrophies and myopathies, heart and great vessel malformations, cardiomyopathies, conduction disorders, dysrhythmias, respiratory malformations, chronic respiratory disease, cystic fibrosis, congenital anomalies, chronic renal failure, congenital anomalies, chronic liver disease and cirrhosis, inflammatory bowel disease, sickle cell disease, hereditary anemias, hereditary immunodeficiency, acquired immunodeficiency, amino acid metabolism, carbohydrate metabolism, lipid metabolism, storage disorders, other metabolic disorders, chromosomal anomalies, bone and joint anomalies, diaphragm and abdominal wall, other congenital anomalies, malignant neoplasms
Castilho et al, 2019	Presence of 2+ included non-communicable diseases	10	DM, CAD (including MI and ischaemia), CKD, cerebrovascular disease (cerebrovascular events and TIA), highgrade hyperlipidemia, venous thromboembolism, cirrhosis, osteonecrosis, osteoporosis (osteoporosis, osteopenia and fragility fractures), all non-AIDS-defining cancers
Castilho et al, 2020	Presence of 2+ included non-communicable diseases	11	HTN, DM, CAD (including MI, CAD, and ischemic heart disease), stage 3-5 CKD, PVD, cerebrovascular disease, chronic liver disease (cirrhosis, hepatic steatosis, and chronic liver disease), hyperlipidemia, obesity, dementia, non–AIDS-defining cancers (excluding basal and squamous cell skin cancers)
Chan et al, 2019	Presence of 2+ included chronic diseases	30	DM, stroke or TIA, CAD, HF, AFIB, COPD or asthma, stage 4-5 CKD, DEP (ever diagnosed + SSRI in the previous year), alcohol problems, anxiety, bronchiectasis, cancer (ever diagnosed), chronic liver disease, dementia or Alzheimer's disease, diverticulitis of the intestine, epilepsy, glaucoma, hypothyroidism, IBD, learning disabilities, motor neurone disease, MS, osteoarthritis (active prescription) or chronic severe back pain, osteoporosis, Parkinson's disease, peripheral arterial cisease, prostate disorders, psoriasis, psychoactive substance misuse, RA, severe mental illness
Demirchyan et al, 2013	Presence 2+ of included conditions	Not specified	Not specified

Dhalwani et al, 2017	Presence of 2+ included chronic diseases	18	HTN, DM, stroke, CHF, MI, COPD, AR, angina, lung disease, asthma, osteoporosis, cancer, hearing problems, Parkinson's disease, Alzheimer's disease, dementia, macular degeneration, glaucoma
Dugravot et al, 2020	Presence of 2+ included chronic diseases	9	DM, stroke, CHD, COPD, DEP, AR, cancer, dementia, Parkinson's disease
Freisling et al, 2020	Presence of 2+ included chronic diseases	3	CVD, cancer, DM
Guaraldi et al, 2015	Presence of 2+ included chronic diseases	8	HTN, DM, CVD, COPD, CKD, cirrhosis, osteoporosis, any cancer
Haug et al, 2020	Not specified	130	DM, intestinal infectious diseases, tuberculosis, certain, zoonotic bacterial diseases, other bacterial diseases, infections with a predominantly sexual mode of transmission, other spirochaetal diseases, other diseases caused by chlamydiae, rickettsioses, viral infections of the central nervous system, arthropod-borne viral fevers and viral haemorrhagic fevers, viral infections characterized by skin and mucous membrane lesions, viral hepatitis, human immunodeficiency virus (HIV) disease, other viral diseases, mycoses, protozoal diseases, helminthiases, pediculosis, acariasis and other infestations, sequelae of infectious and parasitic diseases, bacterial, viral and other infectious agents, other infectious diseases, malignant neoplasms, in-situ neoplasms, benign neoplasms, neoplasms of uncertain or unknown behaviour, nutritional anaemias, haemolytic anaemias, aplastic and other anaemias, coagulation defects, purpura and other haemorrhagic conditions, other diseases of blood and blood-forming organs, certain disorders involving the immune mechanism, disorders of thyroid gland, other disorders of glucose regulation and pancreatic internal secretion, disorders of other endocrine glands, malnutrition, other nutritional deficiencies, obesity and other hyperalimentation, metabolic disorders, organic, including symptomatic, mental disorders, mental and behavioral disorders due to psychoactive substance use, schizophrenia, schizotypal and delusional disorders, mood (affective) disorders, neurotic, stress-related and somatoform disorders, behavioral syndromes associated with physiological disturbances and physical factors, disorders of adult personality and behaviour, mental retardation, disorders of psychological development, behavioral and emotional disorders with onset usually occurring in childhood and adolescence, unspecified mental disorder, inflammatory diseases of the central nervous system, systemic atrophies primarily affecting the central nervous system, extrapyramidal and movement disorders, other degenerative diseases of the nervous system, demyelinating diseases of the central nervous system, episodic and paroxysmal disorders, nerve, nerve root and plexus disorders, polyneuropathies and other disorders of the peripheral nervous system, diseases of myoneural junction and muscle, cerebral palsy and other paralytic syndromes, other disorders of the nervous system, disorders of eyelid, lacrimal system and orbit, disorders of conjunctiva, disorders of sclera, cornea, iris and ciliary body, disorders of lens, disorders of choroid and retina, glaucoma, disorders of vitreous body and globe, disorders of optic nerve and visual pathways, disorders of ocular muscles, binocular movement, accommodation and refraction, visual disturbances and blindness, other disorders of eye and adnexa, diseases of external ear, diseases of middle ear and mastoid, diseases of inner ear, other disorders of ear, acute rheumatic fever, chronic rheumatic

			heart diseases, hypertensive diseases, ischaemic heart diseases, pulmonary heart disease and diseases of pulmonary circulation, other forms of heart disease, cerebrovascular diseases, diseases of arteries, arterioles and capillaries, diseases of veins, lymphatic vessels and lymph nodes, not elsewhere classified, other and unspecified disorders of the circulatory system, acute upper respiratory infections, influenza and pneumonia, other acute lower respiratory infections, other diseases of upper respiratory tract, chronic lower respiratory diseases, lung diseases due to external agents, other respiratory diseases principally affecting the interstitium, suppurative and necrotic conditions of lower respiratory tract, other diseases of pleura, other diseases of the respiratory system, diseases of oral cavity, salivary glands and jaws, diseases of oesophagus, stomach and duodenum, diseases of appendix, hernia, noninfective enteritis and colitis, other diseases of intestines, diseases of peritoneum, diseases of liver, disorders of gallbladder, biliary tract and pancreas, other diseases of the digestive system, infections of the skin and subcutaneous tissue, bullous disorders, dermatitis and eczema, papulosquamous disorders, urticaria and erythema, radiation-related disorders of the skin and subcutaneous tissue, disorders of skin appendages, other disorders of the skin and subcutaneous tissue, arthropathies, systemic connective tissue disorders, dorsopathies, soft tissue disorders, osteopathies and chondropathies, other disorders of the musculoskeletal system and connective tissue, glomerular diseases, renal tubulo-interstitial diseases, renal failure urolithiasis, other disorders of kidney and ureter, other diseases of urinary system, diseases of male genital organs, disorders of breast, inflammatory diseases of female pelvic organs, noninflammatory disorders of female genital tract, other disorders of the genitourinary system
Hohmann et al, 2019	Co-occurrence of 2+ included chronic diseases in respiratory multimorbidity	2	Rhinitis, asthma
Hussin et al, 2018	Co-occurrence of 2+ included chronic diseases	15	HTN, DM, stroke, COPD, hyperlipidemia, osteoarthritis, heart diseases, cataract/glaucoma, renal failure, asthma, tuberculosis, gout, hip fracture, thyroid disorders, cancer
Jørgensen et al, 2020	Not specified	506	Essential (primary) HTN, secondary HTN, insulin-dependent DM, non-insulin-dependent DM, unspecified DM, stroke, acute MI, AFIB and flutter, HF, other bacterial intestinal infections, viral and other specified intestinal infections, diarrhoea and gastroenteritis of presumed infectious origin, other septicaemia, erysipelas, bacterial infection of unspecified site, zoster (herpes zoster), chronic viral hepatitis, viral infection of unspecified site, candidiasis, malignant neoplasm of colon, malignant neoplasm of rectum, malignant neoplasm of bronchus and lung, malignant melanoma of skin, other malignant neoplasms of skin, malignant neoplasm of breast, malignant neoplasm of cervix uteri, malignant neoplasm of corpus uteri, malignant neoplasm of ovary, malignant neoplasm of prostate, malignant neoplasm of bladder, secondary and unspecified malignant neoplasm of lymph nodes, secondary malignant neoplasm of respiratory and digestive organs, secondary malignant neoplasm of other sites, diffuse non-Hodgkin's lymphoma, other and unspecified types of non-Hodgkin's lymphoma, lymphoid leukaemia, melanoma in situ, carcinoma in situ of other and unspecified genital organs, benign neoplasm of mouth and pharynx, benign neoplasm of major salivary glands, benign neoplasm of colon, benign neoplasm of other and ill-defined parts of digestive system, benign neoplasm of middle ear and respiratory system, benign lipomatous neoplasm, haemangioma and lymphangioma, other benign neoplasms of connective and other soft tissue, other benign neoplasms of skin, benign neoplasm of breast, leiomyoma of uterus, benign neoplasm of ovary, benign neoplasm of urinary organs, benign neoplasm of meninges, neoplasm of uncertain or unknown behaviour of oral cavity and

			digestive organs, polycythaemia vera, myelodysplastic syndromes, other neoplasms of uncertain or unknown behaviour of lymphoid, neoplasm of uncertain or unknown behaviour of other and unspecified sites, iron deficiency anaemia, vitamin B12 deficiency anaemia, acute post-haemorrhagic anaemia, anaemia in chronic diseases classified elsewhere, other anaemias, other coagulation defects, purpura and other haemorrhagic conditions, other diseases of blood and blood-forming organs, other hypothyroidism, other nontoxic goitre, thyrotoxicosis (hyperthyroidism), thyroiditis, other disorders of pancreatic internal secretion, hyperparathyroidism and other disorders of parathyroid gland, other disorders of adrenal gland, vitamin A deficiency, deficiency of other B group vitamins, vitamin D deficiency, other vitamin deficiencies, sequelae of malnutrition and other nutritional deficiencies, obesity, disorders of lipoprotein metabolism and other lipidaemias, disorders of mineral metabolism, volume depletion, other disorders of fluid, other metabolic disorders, postprocedural endocrine and metabolic disorders, unspecified dementia, organic amnesic syndrome, delirium, other mental disorders due to brain damage and dysfunction and to physical disease, mental and behavioural disorders due to use of alcohol, mental and behavioural disorders due to use of tobacco, schizophrenia, persistent delusional disorders, acute and transient psychotic disorders, schizoaffective disorders, manic episode. bipolar affective disorder, depressive episode, recurrent depressive disorder, persistent mood (affective) disorders, unspecified mood (affective) disorder, other anxiety disorders, reaction to severe stress, somatoform disorders, specific personality disorders, mild mental retardation, moderate mental retardation, severe mental retardation, unspecified mental retardation, mental disorder, Parkinson's disease, other extrapyramidal and movement disorders, other degenerative diseases of nervous system, epilepsy, status epilepticus, migraine, other headache syndromes, transient cerebral ischaemic attacks and related syndromes, sleep disorders, disorders of trigeminal nerve, nerve root and plexus disorders, mononeuropathies of upper limb, other polyneuropathies, other disorders of peripheral nervous system, disorders of autonomic nervous system, hydrocephalus, other disorders of brain, other disorders of central nervous system, other disorders of nervous system, other inflammation of eyelid, other disorders of eyelid, disorders of lacrimal system, disorders of orbit, conjunctivitis, other disorders of conjunctiva, keratitis, other disorders of cornea, senile cataract, other cataract, other disorders of lens, retinal detachments and breaks, retinal vascular occlusions, other retinal disorders, retinal disorders in diseases classified elsewhere, glaucoma, disorders of vitreous body, disorders of globe, other disorders of optic (2nd) nerve and visual pathways, other strabismus, disorders of refraction and accommodation, visual disturbances, blindness and low vision, other disorders of eye and adnexa, otitis externa, other disorders of external ear, nonsuppurative otitis media, suppurative and unspecified otitis media, cholesteatoma of middle ear, perforation of tympanic membrane, otosclerosis, disorders of vestibular function, vertiginous syndromes in diseases classified elsewhere, other diseases of inner ear, conductive and sensorineural hearing loss, other hearing loss, other disorders of ear, other disorders of ear in diseases classified elsewhere, hypertensive heart disease, hypertensive renal disease, angina pectoris, other acute ischaemic heart diseases, chronic ischaemic heart disease, pulmonary embolism, other pulmonary heart diseases, nonrheumatic mitral valve disorders, nonrheumatic aortic valve disorders, cardiomyopathy, atrioventricular and left bundle-branch block, other conduction disorders, cardiac arrest, paroxysmal tachycardia, other cardiac arrhythmia, complications and ill-defined descriptions of heart disease, subarachnoid haemorrhage, intracerebral haemorrhage, cerebral infarction, occlusion and stenosis of precerebral arteries, other cerebrovascular diseases, sequelae of cerebrovascular disease, atherosclerosis, aortic aneurysm and dissection, other aneurysm, other PVDs, other disorders of arteries and arterioles, diseases of capillaries, phlebitis and thrombophlebitis, other venous embolism and thrombosis, varicose veins of lower extremities, haemorrhoids, other disorders of veins, nonspecific, lymphadenitis, other noninfective disorders of lymphatic vessels and lymph nodes, hypotension, other and unspecified disorders of circulatory system, influenza, bacterial pneumonia, pneumonia, acute bronchitis, unspecified acute lower respiratory infection, chronic sinusitis, nasal polyp, other disorders of nose and nasal sinuses, chronic diseases of tonsils and adenoids, chronic laryngitis and laryngotracheitis, diseases of vocal cords and larynx, bronchitis, simple and mucopurulent chronic bronchitis, unspecified

			chronic bronchitis, emphysema, other COPD, asthma, bronchiectasis, pulmonary oedema, other interstitial pulmonary diseases, pyothorax, pleural effusion, pleural plaque, pneumothorax, respiratory failure, other respiratory disorders, dental caries, diseases of pulp and periapical tissues, gingivitis and periodontal diseases, dentofacial anomalies (including malocclusion),other disorders of teeth and supporting structures, cysts of oral region, other diseases of jaws, diseases of salivary glands, other diseases of lip and oral mucosa, oesophagitis, gastro-oesophageal reflux disease, other diseases of oesophagus, gastric ulcer, duodenal ulcer, peptic ulcer, gastritis and duodenitis, dyspepsia, other diseases of stomach and duodenum, acute appendicitis, inguinal hernia, femoral hernia, umbilical hernia, ventral hernia, diaphragmatic hernia, Crohn's disease (regional enteritis),ulcerative colitis, other noninfective gastroenteritis and colitis, vascular disorders of intestine, paralytic ileus and intestinal obstruction without hernia, diverticular disease of intestine, irritable bowel syndrome, other functional intestinal disorders, fissure and fistula of anal and rectal regions, abscess of anal and rectal regions, other diseases of anus and rectum, other diseases of intestine, peritonitis, alcoholic liver disease, chronic hepatitis, fibrosis and cirrhosis of liver, other diseases of liver, cholelithiasis, cholecystitis, other diseases of biliary tract, acute pancreatitis, other diseases of pancreas, postprocedural disorders of digestive system, other diseases of digestive system, cutaneous abscess, cellulitis, other local infections of skin and subcutaneous tissue, pruritus, other dermatitis, psoriasis, lichen planus, urticaria, skin changes due to chronic exposure to nonionizing radiation, nail disorders, rosacea, follicular cysts of skin and subcutaneous tissue, seborrhoeic keratosis, decubitus ulcer, atrophic disorders of skin, ulcer of lower limb, other disorders of skin and subcutaneous tissue, pyogenic arthritis, reactive arthropathies, seropositive rheumatoid arthritis, other rheumatoid arthritis, psoriatic and enteropathic arthropathies, gout, other crystal arthropathies, other arthritis, polyarthrosis, coxarthrosis (arthrosis of hip), gonarthrosis (arthrosis of knee), arthrosis of first carpometacarpal joint, other arthrosis, acquired deformities of fingers and toes, other acquired deformities of limbs, internal derangement of knee, other specific joint derangements, other joint disorders, other necrotizing vasculopathies, other systemic involvement of connective tissue, scoliosis, other deforming dorsopathies, other inflammatory spondylopathies, spondylosis, other spondylopathies, cervical disc disorders, other intervertebral disc disorders, other dorsopathies, dorsalgia, other disorders of muscle, synovitis and tenosynovitis, spontaneous rupture of synovium and tendon, other disorders of synovium and tendon, soft tissue disorders related to use, other bursopathies, fibroblastic disorders, shoulder lesions, enthesopathies of lower limb, other enthesopathies, other soft tissue disorders, osteoporosis with pathological fracture, osteoporosis without pathological fracture, other disorders of bone density and structure, osteomyelitis, other disorders of bone, postprocedural musculoskeletal disorders, biomechanical lesions, recurrent and persistent haematuria, acute tubulo-interstitial nephritis, obstructive and reflux uropathy, acute renal failure, chronic renal failure, unspecified renal failure, calculus of kidney and ureter, calculus of lower urinary tract, other disorders of kidney and ureter, cystitis, neuromuscular dysfunction of bladder, other disorders of bladder, urethral stricture, other disorders of urethra, other disorders of urinary system, hyperplasia of prostate, hydrocele and spermatocele, orchitis and epididymitis, redundant prepuce, other disorders of penis, other disorders of male genital organs, benign mammary dysplasia, hypertrophy of breast, unspecified lump in breast, other disorders of breast, other inflammation of vagina and vulva, female genital prolapse, noninflammatory disorders of ovary, polyp of female genital tract, other noninflammatory disorders of uterus, dysplasia of cervix uteri, other noninflammatory disorders of vagina, other noninflammatory disorders of vulva and perineum, excessive, pain and other conditions associated with female genital organs and menstrual cycle, menopausal and other perimenopausal disorder, postprocedural disorders of genitourinary system, congenital malformations of cardiac septa, Down's syndrome, congenital deformities of feet, abnormalities of heart beat, cardiac murmurs and other cardiac sounds, haemorrhage from respiratory passages, cough, abnormalities of breathing, pain in throat and chest, other symptoms and signs involving the circulatory and respiratory systems, abdominal and pelvic pain, nausea and vomiting, dysphagia, faecal incontinence, unspecified jaundice, other symptoms and signs involving the digestive system and abdomen, rash and other nonspecific skin eruption, localized swelling, abnormal involuntary movements, abnormalities

			of gait and mobility, other symptoms and signs involving the nervous and musculoskeletal systems, unspecified haematuria, unspecified urinary incontinence, retention of urine, polyuria, other symptoms and signs involving the urinary system, somnolenc, other symptoms and signs involving cognitive functions and awareness, dizziness and giddiness, other symptoms and signs involving general sensations and perceptions, voice disturbances, fever of other and unknown origin, headache, pain, malaise and fatigue, syncope and collapse, convulsions, enlarged lymph nodes, oedema, symptoms and signs concerning food and fluid intake, unknown and unspecified causes of morbidity, elevated erythrocyte sedimentation rate and abnormality of plasma viscosity, elevated blood glucose level, abnormal serum enzyme levels, other abnormal findings of blood chemistry, abnormal findings on diagnostic imaging of lung, abnormal findings on diagnostic imaging of other body structures, abnormal results of function studies, superficial injury of head, open wound of head, fracture of skull and facial bones, injury of eye and orbit, intracranial injury, other and unspecified injuries of head, superficial injury of neck, dislocation, superficial injury of thorax, fracture of rib(s), injury of other and unspecified intrathoracic organ, superficial injury of abdomen, fracture of lumbar spine and pelvis, superficial injury of shoulder and upper arm, fracture of shoulder and upper arm, dislocation, injury of muscle and tendon at shoulder and upper arm level, superficial injury of forearm, open wound of forearm, fracture of forearm, dislocation, superficial injury of wrist and hand, open wound of wrist and hand, fracture at wrist and hand level, dislocation, injury of muscle and tendon at wrist and hand level, superficial injury of hip and thigh, open wound of hip and thigh, fracture of femur, dislocation, superficial injury of lower leg, open wound of lower leg, fracture of lower leg, dislocation, injury of muscle and tendon at lower leg level, superficial injury of ankle and foot, open wound of ankle and foot, fracture of foot, dislocation, superficial injuries involving multiple body regions, unspecified multiple injuries, other injuries of lower limb, injury of unspecified body region, foreign body on external eye, foreign body in ear, foreign body in respiratory tract, foreign body in alimentary tract, poisoning by systemic antibiotics, poisoning by nonopioid analgesics, poisoning by narcotics and psychodysleptics (hallucinogens), poisoning by antiepileptic, poisoning by psychotropic drugs, poisoning by primarily systemic and haematological agents, poisoning by agents primarily affecting the cardiovascular system, poisoning by diuretics and other and unspecified drugs, toxic effect of alcohol, toxic effect of contact with venomous animals, toxic effect of other and unspecified substances, adverse effects, certain early complications of trauma, complications following infusion, complications of procedures, complications of genitourinary prosthetic devices, complications of internal orthopaedic prosthetic devices, complications of other internal prosthetic devices, other complications of surgical and medical care, sequelae of injuries of upper limb, sequelae of injuries of lower limb, sequelae of other and unspecified effects of external causes
Kivimäki et al, 2017	Developing 2+ included chronic diseases	3	DM, stroke, CHD
Lau et al, 2019	Presence of 2+ included chronic diseases	6	HTN, DM, stroke, cancer, heart diseases, chronic lung diseases
Licher et al, 2019	Presence of 2+ included chronic diseases	6	DM, stroke, COPD, heart disease, cancer, neurodegenerative disease
Marengoni et al, 2018	Not specified	17	HTN/increased blood pressure, HF, DM, DEP, rheumatoid AR, degenerative AR of the back/other illness of the back, angina pectoris, cancer, asthma, pulmonary emphysema/chronic bronchitis, gallstones/gall bladder inflammation, other articular disease, chronic urethritis/nephritis, cerebrovascular disease, coronary bypass, angioplasty, other psychological illness

Melis et al, 2014	Co-occurrence of 2+ included chronic diseases	39	HTN, DM, HF, AFIB, COPD, DEP, AR, alcohol dependence syndrome, anemia, rheumatoid AR, cancer, cardiomiopathy, cerebrovascular diseases, cholelithiasis, calculus of kidney/ureter, colitis, chronic pulmonary heart disease, chronic renal failure, chronic rheumatic heart disease, crystal arthropathies, deafness, diverticula of intestine, liver diseases, pancreas diseases, epilepsy, disorders of the eyes, functional digestive disorders, hip fracture, hyperplasia of prostate, ischemic heart disease, intestinal malabsorption, lung diseases due to external agents, migraine, Parkinson disease, peripheral nerve system diseases, schizophrenia, thyroid disorders, dementia, rheumatic polymyalgia
Mounce et al, 2018	Presence of 2+ included chronic diseases	15	HTN, DM, stroke, chronic HF, COPD, AR, asthma, osteoporosis, heart arrhythmia, cancer, ischemic heart disease, Parkinson's disease, 1 or more affective mental health conditions (DEP, anxiety, or emotional problems), 1 or more psychotic mental health conditions (schizophrenia, psychosis, bipolar disorder, or hallucinations), dementia (including Alzheimer's disease)
Niedzwiedz et al, 2019	Presence of 2+ included chronic diseases	8	High blood pressure; DM (or high blood sugar); stroke; cancer or a malignant tumour (excluding minor skin cancer); chronic lung disease; heart attack, CHD, angina, congestive HF, or other heart problems; emotional, nervous, or psychiatric problems; AR (or rheumatism)
Ryan et al, 2018	Presence 2+ of included conditions	16	HTN, DM, AR, cardiac conditions, cerebrovascular disease, high cholesterol, chronic respiratory disease, liver disease, eye disease, cognitive impairment, osteoporosis, cancer, Parkinson's disease, emotional/psychological condition, stomach ulcers, varicose veins
Salloum et al, 2018	Not specified	26	Blood and lymphatic system disorders, cardiac disorders, congenital, familial and genetic disorders, ear and labyrinth disorders, endocrine disorders, eye disorders, gastrointestinal disorders, general disorders and administration site conditions, hepatobiliary disorders, immune system disorders, infections and infestations, injury, poisoning and procedural complications, investigations, metabolism and nutrition disorders, musculoskeletal and connective tissue disorders, neoplasms benign, malignant and unspecified (including cysts and polyps), nervous system disorders, pregnancy, puerperium and perinatal conditions, psychiatric disorders, renal and urinary disorders, reproductive system and breast disorders, respiratory, thoracic and mediastinal disorders, skin and subcutaneous tissue disorders, social circumstances, surgical and medical procedures, vascular disorders
Singh-Manoux et al, 2018	Presence of 2+ included chronic diseases in cardiometabolic multimorbidity	3	DM, stroke, CHD
St. Sauver et al, 2015	Presence of 2+ included chronic diseases	20	HTN, DM, stroke, CAD, congestive HF, COPD, CKD, DEP, AR, cardiac arrhythmias, hyperlipidemia, asthma, autism spectrum disorder, cancer, dementia (including alzheimer's and other senile dementias), hepatitis, human immunodeficiency virus, osteoporosis, schizophrenia, substance abuse disorder (drug and alcohol)
Strauss et al, 2014	Presence of 2+ included chronic diseases	42	DM, congestive HF, AFIB, AR, benign neoplasm of prostate, carcinoma in-situ, neurofibromatosis, Von Recklinghausen's disease, hypothyroidism, pure hypercholesterolaemia, obesity hereditary haemolytic anaemia, senile/presenile, dementia, hereditary and idiopathic peripheral neuropathy, primary open-angle glaucoma, senile cataract, corneal opacity and disorders of cornea, deafness, mitral stenosis, rheumatic heart disease, high blood pressure, hypertensive heart disease, hypertensive renal disease, angina pectoris, ischaemic heart disease,

			cerebral atherosclerosis, cerebrovascular disease, intermittent claudication, chronic bronchitis, emphysema, extrinsic allergic alveolitis, pulmonary oedema, diffuse pulmonary fibrosis, nephritis and nephropathy, hydronephrosis, prostatism, rheumatoid, osteoarthritis, arthropathy, cervical spondylosis, Paget's disease of bone, osteoporosis, late effect - nervous system injury, war injuries
Tomasdottir et al, 2016	Co-occurrence of 2+ included chronic diseases	17	HTN, DM, CVD, hyperlipidaemia, obesity, chronic back or neck pain, thyroid disease, asthma, cancer, mental health problems, epilepsy, gastro-oesophageal reflux, ankylosing spondylitis, osteoarthrosis, rhematoid AR, fibromyalgia, osteoporosis
Ungprasert et al, 2017	Presence of 2+ included chronic diseases	18	HTN, DM, congestive HF, CHD, COPD, CKD, DEP, AR (rhematoid or osteoarthritis), arrhythmia, hyperlipidemia, cerebrovascular (stroke or transient ischemic attack), asthma, cancer, dementia/alzheimer's disease, hepatitis, major osteoporotic fracture, schizophrenia, substance abuse disorder
van den Akker et al, 1998	Co-occurrence of 2+ included chronic diseases	335	All active International Classification of Primary Care (ICPC) codes, excluding signs & symptoms
Wikström et al, 2015	Co-occurrence of 2+ included chronic diseases	5	DM, CVD, COPD/asthma, RA, cancer
Wilson-Genderson et al, 2017	Presence of 2+ included chronic diseases	5	HTN, DM, AR, heart disease, pulmonary disease
Xu et al, 2018	Presence of 2+ included chronic diseases	3	DM, stroke, CHD
Xu et al, 2019	Presence of 2+ included chronic diseases	9	HTN, DM, CVD, COPD, AR, mental health disorders (DEP/anxiety), asthma, osteoporosis, cancer
Xu, Mishra & Jones, 2019	Presence of 2+ included chronic diseases	9	HTN, DM, stroke, COPD, AR, heart disease, cancer, asthma, osteoporosis
Xu et al, 2020	Presence of 2+ included chronic diseases	11	HTN, DM, stroke, COPD, DEP, AR, heart disease, osteoporosis, asthma, anxiety, breast cancer